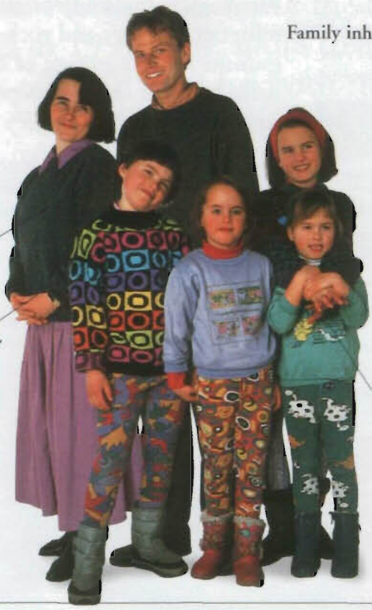
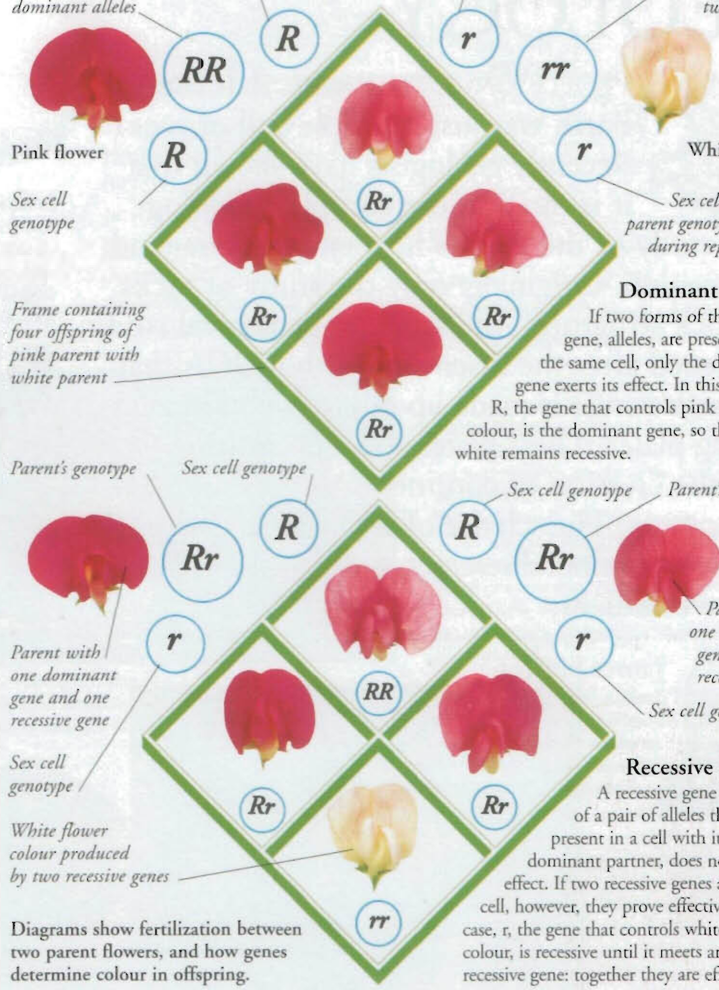
Heredity

Heredity is the transmission of characteristics from one generation to the next. These inherited characteristics, such as size, shape, and colour, are determined by genes passed on by parents. When different forms of the same genetic characteristic meet (for example, blue and brown eye colour), some genes are dominant (effective) and some are recessive (ineffective).



Family inheritance



Dominant genes

If two forms of the same gene, alleles, are present in the same cell, only the dominant gene exerts its effect. In this case, R, the gene that controls pink flower colour, is the dominant gene, so the colour white remains recessive.

Recessive genes

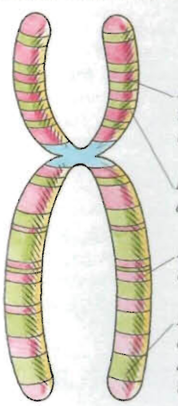
A recessive gene is one of a pair of alleles that, if present in a cell with its dominant partner, does not exert its effect. If two recessive genes are in the cell, however, they prove effective. In this case, r, the gene that controls white flower colour, is recessive until it meets another recessive gene: together they are effective.

Diagrams show fertilization between two parent flowers, and how genes determine colour in offspring.

Sex chromosomes

Whether an animal is male or female is determined by one pair of chromosomes called the sex chromosomes. In humans and other mammals, a female's sex chromosomes are identical and are called XX. In males, one chromosome is smaller, and the pair is called XY.

Human X chromosome



Sex determination

Sperm and eggs each carry one sex chromosome. Eggs carry an X chromosome, and sperm carry an X or a Y. When a sperm and an egg meet during fertilization, there is a 50:50 chance of producing a male (XY) or a female (XX), depending on which chromosome the sperm is carrying.

Most ginger cats are male.

Tortoise-shell cats are always female.

Sex-linked inheritance

Sex chromosomes also carry genes that determine other characteristics apart from an animal's sex. More of these sex-linked genes are found on X chromosomes than Y. So some characteristics are specific only to males or to females.

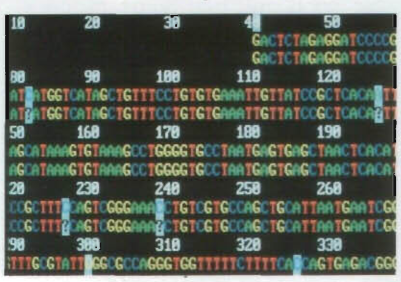
Genetic code

DNA contains the instructions to make the proteins that construct the cell and control its functions. Four chemical bases – adenine (A), cytosine (C), guanine (G), and thymine (T) – combine in pairs to form a sequence, or code. The cell then translates this code and produces a protein.

Human Genome Project

The genome is the complete set of genes found in the nucleus of every body cell. In the 1980s the Human Genome Project set out to identify all of those genes by working out the sequence of bases. In 2001 they announced the existence of some 30,000 genes.

Genetic codes are made up of bases (A, C, G, T).



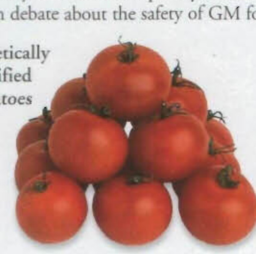
Genetic engineering

Genetic engineering involves taking genes from one cell and inserting them into another cell. This gives the cell new characteristics, which are determined by the transferred gene. In the future, genetic engineering may be used on human cells, so that genetic disorders can be eliminated.

Genetically modified (GM) food

Food products can also be altered by genetic modification or engineering. For example, scientists can genetically engineer certain fruits and vegetables, so that they do not rot so quickly. There is much debate about the safety of GM foods.

Genetically modified tomatoes



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